

CAPITAL ALLOCATION

Full Learning & Calculation Guide

8 screens • 10/10 work packages • deterministic ledger • Monte Carlo • reverse solver • marginal allocation • property • scenarios • attribution

2026/27 UK • Scotland + Rest of UK

10,000 paths default

Audit-oriented

目的：不是教你按哪個按鈕，而是讓你能對每一個數字問「它從哪裡來？這個假設合理嗎？如果不合理，該改哪個 engine / parameter？」

Based on current main • 15 Sep 2026

怎麼使用這份 PDF

第一遍：看「模型在回答什麼」

先不要糾結所有欄位。每個 screen 只抓：Question → Engine → Main output。

第二遍：追 calculation

看到結果時，沿著公式回推：Profile input → annual ledger → simulation/search → view metric。

第三遍：做 audit

把 UI 的數字跟 deck 裡 release benchmark / identities 對照。對不上，就定位到 input、label、engine 或 assumption。

這份 deck 刻意把 4 種「看起來都像預測」的東西分開：

類型	它到底是什麼	你該怎麼讀
Deterministic	每年都用設定的平均報酬	reference / debugging，不是安全性
Monte Carlo	10,000 條市場路徑的統計	概率估計，有 sampling error
Sensitivity	改一個 assumption 再重跑	「如果 X 不同會怎樣」
Stress	人工指定極端路徑	脆弱性測試，不是發生機率

核心原則：不要把 UI 上任何單一數字當成「預言」。它們是 decision-support outputs。

全產品地圖： 8 個畫面，現在全部接 real engine

Overview

現在的資產 / 現金流 + 年度 ledger

FIRE & Monte Carlo

成功機率、失敗、分位數、sequence risk

FIRE Age Curve

不同退休年齡的成功機率曲線

Marginal Allocation

下一筆錢放哪裡最好

Reverse Solver

要多少薪水 / 存款 / 延後幾年才達標

Scenario Comparison

薪資 $\times$ 支出 $\times$ 策略 + named scenarios

Property & Leverage

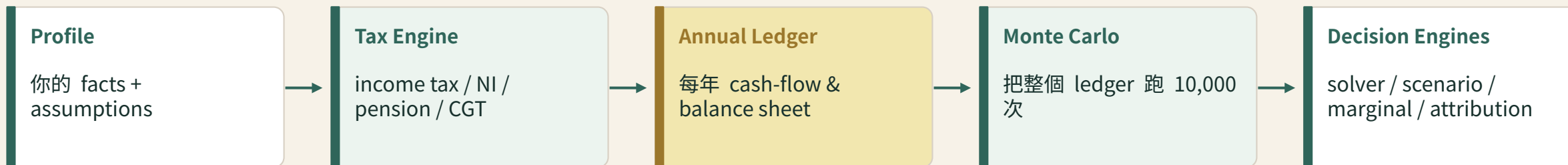
買房、房貸、出租、槓桿、rent-vs-invest

Where It Comes From

哪個 lever 最重要 + sensitivity + stress

所有畫面共用同一個 Profile、同一套 tax / ledger / market assumptions；差別在「怎麼呼叫 engine」與「怎麼比較結果」。

模型不是 8 個 calculator：它是一個 shared engine graph



Profile → Tax → Ledger(year 31...95) → one complete lifetime path

Deterministic = one mean path; Monte Carlo = many generated market paths.

重要： tax 不是獨立顯示數字；它會改可投資 surplus、withdrawal gross-up、GIA disposal、pension drawdown。

重要： Property 不是「房價另外算」； mortgage / housing costs / sale proceeds 都進同一個年度 funding order。

重要： Solver / Scenario 不是用簡化比例推估，而是反覆重跑完整 lifetime simulation。

一條 lifetime path 到底怎麼走？



年齡階段

Accumulation

age < FIRE

Bridge

FIRE ≤ age < pension access

Retirement

age ≥ pension access

$$\text{Nominal salary}_t = \text{salary}_{\text{today}} \times (1 + \text{real salary growth})^t \times \text{inflation index}_t$$

所以 salary growth 是「real」；nominal salary 同時長 real growth + inflation。

為什麼 event order 很重要？

- opening balance 先決定 taxable cash interest / GIA dividend，避免稅與提款互相循環。
- spending / tax 先發生，market return 再作用在 post-flow balance；不是先漲完再花。
- 每一個 Monte Carlo path 都遵守完全相同的 accounting order。

預設 example profile：所有 release benchmark 的基準

項目	預設值	項目	預設值
Current / FIRE / end age	31 / 45 / 95	Target success	90%
Gross salary	£55,000	Real salary growth	3%
Current / retirement spend	£1,650 / mo	Scenario spend	£1,300 / £1,650 / £2,000
Cash / ISA / GIA	£10k / £50k / £15k	Pension	£25k
Pension employee / employer	5% / 5%	Pension access	57
Portfolio	85% eq / 10% bond / 5% cash	Simulation	10,000 paths; seed 421337
Eq mean / vol	7% / 16%	Bond mean / vol	3% / 6%
Cash / property mean	2.5% / 3.5%	Inflation mean	2.5%
Reference withdrawal rate	3.5%	Property	None

關鍵：這些是「example assumptions」，不是 forecasts。你改任何一個，所有 downstream screens 都會重算。

左側輸入很多，先把它們分成 12 組

Personal & target

age, FIRE age, end age, target probability

Income

salary, bonus, real growth, post-FIRE income, state pension

Household

adults, children

Spending

essential, discretionary, phases, floor/comfort, scenarios

Assets

cash, ISA, GIA value/basis/losses, pension, SIPP

Pension policy

method, employee/employer, match, access age, carry-forward

ISA & GIA

ISA used, dividend yield, turnover, gain realisation

Liquidity

emergency months, minimum liquid years, capital needs

Portfolios

ISA/GIA/pension eq-bond-cash weights

Market

mean/vol + 5×5 log-shock correlation

Simulation

count, seed, reference withdrawal rate

Property

value/debt/rates/costs/rent/purchase/sale/refinance

最重要的 mental model：不是所有欄位都一樣「真」

A. 你的 facts

salary、現有資產、pension rate、支出、房貸餘額、年齡。
錯了 = 模型輸入錯。

B. 你的 planning choices

FIRE age、target success、spending case、withdrawal order、planned property。
不是預測，是你要測的 plan。

C. Model assumptions

equity mean/vol、inflation、property growth、correlation。
這些最值得 sensitivity / stress。

D. Technical controls

simulation count、seed、solver bound / precision、preview/full count。
影響計算成本與 sampling，不應被誤讀為財務假設。

E. Tax-policy freeze

2026/27 structure 在未來年度按 inflation 做 constant-real scaling。
這是 disclosure，不是 tax forecast。

Audit 時先問：這個怪結果，是 fact 錯、plan choice、assumption，還是 technical setting？

Tax engine：不是簡單「salary × tax rate」

類別	2026/27 model 支援重點
Personal allowance	£12,570；adjusted net income > £100k 後，每 £2 減 £1
Scottish income tax	non-savings/non-dividend 用 Scotland bands；savings/dividends 用 UK rules
Savings	starting rate + PSA (£1,000 / £500 / £0)；仍佔 band
Dividends	£500 nil-rate band；其餘按 dividend rates
Pension	salary sacrifice / net pay / relief at source 三套 mechanics
Annual allowance	£60k；MPAA £10k；taper tests £200k / £260k，min £10k
ISA	£20k annual capacity
CGT	£3k exemption；ordinary investment gains 18% / 24%
NI	annual Class 1 A/C approximation；state pension age 後轉 C

Engine 算「全年最終 liability」，不是 PAYE payslip / monthly NIC rounding。

未支援：student loan、self-employed NIC、benefits/child benefit charge、foreign relief、special codes...

預設 £55k salary 的 tax / pension reference case

£55,000

gross salary

£9,927.05

income tax

£3,055.60

employee NI

£39,267.35

net annual income

£5,500

total pension funding

5% salary sacrifice + 5% employer contribution → pension added = £2,750 + £2,750 = £5,500

Example has no employer match or NI shareback.

£39,267.35 net income – £19,800 annual spending = £19,467.35 before allocation

This is the pre-allocation reference surplus from the tax support doc.

這個 reference 非常好用：如果你沒有改 profile，但 Overview 的 first-year tax / take-home 差很大，應該先查 input / UI mapping，而不是怪 Monte Carlo。

Annual ledger：每年完整 settlement，不是把幾條公式相加

步驟	主要計算	為什麼
1. Opening	cash / ISA / GIA / pension / property / debt	今年的 tax & funding 從這裡開始
2. Investment income	cash interest、GIA dividends、turnover gains	使用 opening balance，避免 circularity
3. Active income	salary、bonus、other income、state pension、pension contrib、tax/NI	每年重新 settle
4. Obligations	spending + capital needs + property costs + withdrawal tax gross-up	依 withdrawal order 找 funding
5. Surplus	先 emergency reserve，再 ISA/GIA / chosen policy	保持 liquidity gate
6. Returns	對 post-flow balances 套 market return	順序決定 compounding
7. Close	closing balance、net worth、failures	下一年的 opening

Failure 不會因為未來資產回來而被「洗掉」：例如 mortgage_shortfall 是 persistent failure record。

核心 accounting identity：你要學會用它抓 bug

financial closing = financial opening + active income + pension added - pension cash paid - total tax - living/capital/property costs - interest/principal + transaction cash flows + financial returns

The engine asserts year-by-year reconciliation.

net-worth change = active income + pension added - pension cash paid - tax - living/capital costs - property operations - accrued mortgage interest - transaction costs + financial returns + property appreciation

Property value and mortgage debt belong to net worth, not “accessible cash” .

最常見錯誤： 把 property equity 當可花用資產；engine 不會，只有 sale 才釋放 equity。

最常見錯誤： 把 unrealised GIA gain 當每年 taxable gain；engine 只對 realised gain / turnover / disposal 做 CGT。

最常見錯誤： pension 鎖住期間拿來補 FIRE bridge；engine 明確禁止。

你看到的「£50k → £54.2k」：先拆 label，而不是猜 tax

£50,000

左側 Gross salary = salaryAnnual

Total gross inflows

Overview 的 grossIncome 本質

≠ salary

UI label 可能讓人誤解

grossIncome = employment income + other income + state pension + taxable cash interest + GIA dividends + gross pension withdrawal + rental income

所以它是 annual gross inflow，而不是「salary 被改掉」。

用 default profile 做 sanity check：

£50,000 salary + (£10,000 × 2.5% cash) + (£15,000 × 2% dividends) ≈ £50,550

沒有其他收入、rent、state pension 或 pension withdrawal 時。

所以 £54,200 值得追

它暗示當下 profile 還有約 £3,650 其他 gross inflow，或 UI / state 有 mapping 問題。建議把 label 改成「Total gross income」並顯示 component breakdown。

Overview

先建立「現在在哪裡」與「單一路徑怎麼算」的共同語言。

01

Overview : 這頁不是 FIRE verdict , 是 reconciliation dashboard

現在有多少?

cash / ISA / GIA / pension / property equity / debt

今年賺多少、花多少?

gross income / tax / NI / pension / spending / investable surplus

FIRE reference 差多遠?

reference FIRE number / accessible / locked / bridge

每年到底發生什麼?

full annual ledger with opening → flows → returns → closing

不要在這頁把「expected-value deterministic path」當成 68.98% 那種 probability。Overview 的價值是透明、可追帳。

Reference FIRE number = annual retirement spending / reference withdrawal rate

Default 3.5% is a reference arithmetic only; not the Monte Carlo safety result.

Overview 的 5 個 FIRE metric：不要混在一起

Metric	意思	讀法
Reference FIRE number	退休年支出 / reference WR	像尺，不是 pass/fail
Investable assets at FIRE	FIRE 當年 financial net worth , real	有多少金融資產
Accessible at FIRE	可提款的 liquid capital	能不能過 bridge
Locked at FIRE	尚未可領 pension	有錢，但現在不能花
Required bridge capital	FIRE → pension access 所需 capital	早退最容易出問題的段落
Required post-pension capital	pension access 後所需 capital	第二段 funding need
Liquid FIRE ratio	accessible / required bridge	< 1 = bridge 壓力
Pension coverage ratio	locked / post-pension need	高也不代表 bridge 安全

真正的 FIRE safety 仍然要回到 Monte Carlo：因為同樣的「資產額」在不同 return sequence 下，結果可能完全不同。

Overview ledger：一列一列怎麼 debug

你看到的欄位	對應 engine 問題
salary / bonus / gross income	收入來源是否正確？ gross income 是否被誤認為 salary ？
income tax / NI / total tax	tax region 、 pension method 、 other income 、 dividend/interest 是否一致？
pension added	employee + employer + match + NI shareback 有沒有 double count ？
spending required / funded	支出 schedule 、 rent removal 、 property cost 是否重複？
withdrawals	cash → GIA → ISA → pension 的 gross-up 是否合理？
allocated ISA/GIA/cash	emergency reserve / allowance 是否 gate 住？
investment return	對 post-flow balance 而非 opening balance
closing / accessible / locked	是否符合 reconciliation identity ？
failure / shortfall	哪個 obligation 第一次 fund 不到？

Debug 順序： first bad year → bad line item → upstream input ； 不要直接從 terminal wealth 反推。

FIRE & Monte Carlo

同一個 lifetime ledger ，改成 10,000 條可能的 market path ，問「計畫有多常成功？」

02

Deterministic vs Monte Carlo：差別只有 market path，不是另一套財務公式

Deterministic

每年使用 configured nominal mean。
適合：reconciliation、reference、debug。
不適合：當安全性 verdict。

Monte Carlo

每條 path 生成不同 equity/bond/cash/property/inflation shocks，再把完整 ledger 跑一次。
適合：success probability、distribution、sequence risk。

Same profile + same tax + same ledger + different market path → path outcome

這是為什麼 Monte Carlo 結果仍然可以逐年 audit。

成功率不是「平均報酬達標機率」，而是 10,000 個完整 lifetime plans 裡，有多少沒有觸發 funding failure。

Monte Carlo market generator : mean/vol 不是直接塞進 Normal

輸入的是 arithmetic annual nominal mean m 與 volatility v ; engine 轉成 lognormal parameters :

$$\sigma = \sqrt{ \ln(1 + (v / (1+m))^2) }$$

$$\mu = \ln(1+m) - \sigma^2 / 2$$

$$\text{return} = \exp(\mu + \sigma \cdot z_{\text{correlated}}) - 1$$

讓 simulated arithmetic mean / volatility 對齊你輸入的設定。

5 個 stochastic drivers

Equities

Bonds

Cash

Property

Inflation

Correlation matrix 是 Gaussian log-growth shocks 的 correlation , 不是 observed arithmetic returns 的 correlation 。這個 distinction 很技術, 但對校準很重要。

Seed、correlation、common random numbers：為什麼 scenario 比較比較公平

Seed

Default 421337。相同 seed + path index → 相同 underlying random stream。

5×5 correlation

Cholesky-style correlation 將 independent shocks 變成 correlated shocks。

Common random numbers

比較兩個 plan 時，path #732 在兩邊遭遇同一組 market shocks。差異比較像是 strategy effect，而不是 luck。

Scenario A(path i) - Scenario B(path i) → paired comparison

這比 A、B 各抽一批不同 random paths 的 noise 小。

所以：不要每個 strategy 用不同 seed 來比較；那會把 market luck 跟 decision effect 混在一起。

Success probability : 什麼叫「一條 path 成功」？

Unfunded essential spending

基本支出 fund 不到

Pre-pension liquidity failure

bridge 期間 accessible assets 不夠

Portfolio depletion

可用金融資產耗盡

Insolvency

總資產 / 義務進入無法維持狀態

Mortgage shortfall

房貸 obligation 未 fund

Success probability = successful lifetime paths / total simulated paths

注意：terminal net worth > 0 並不一定代表過程中從未失敗；engine 保留 failure records。

Percentile、mean、standard error：不要只看一個 headline

Output	公式 / 定義	你在乎什麼
Success probability	$p = \text{successes} / N$	是否高於 target
Sampling SE	$\sqrt{[p(1-p)/N]}$	同一模型因有限 paths 的統計噪音
P10 / Median / P90	wealth distribution quantiles	downside / typical / upside
Mean terminal wealth	$\Sigma \text{ terminal} / N$	會被極端高 wealth 拉高
Worst observed	$\min(\text{path outcome})$	只是樣本最差，不是 theoretical worst
Bridge / depletion probability	specific failures / N	失敗「在哪裡」而不是只看 pass/fail

At $p \approx 0.69$ and $N=10,000$: $SE \approx \sqrt{(0.69 \times 0.31 / 10,000)} \approx 0.46$ percentage points

所以 0.2pp 的差距未必值得宣稱「策略 A 比 B 好」。

Sequence risk diagnostics : 平均報酬一樣，早期 crash 仍可能讓 FIRE 失敗

Early drawdown

FIRE 後前 5 年，real wealth 是否從 peak drawdown > 20%。

Accessible < 2 × spending

可動用 capital 是否跌到不足 2 年 annual spending。

Recovery years

觸發後，需要幾年 real wealth 才回到 trigger peak。

Diagnostic flags

Early equity crash FIRE 前 5 年內任一年 equity return < -20%。

High early inflation window geometric annual inflation > 5%。

這些不是 causal proof

「失敗 path 裡 early crash 很常見」不等於「early crash 是唯一原因」。它是 association / diagnosis。

這頁最值得看的不是 P90，而是：失敗是否集中在 bridge + early retirement sequence。

Release baseline : 預設 plan 的 full-count success = 68.98%

68.98%

10,000-path baseline success

90%

user target

-21.02 pp

gap to target

這個 baseline 的意思不是「你有 68.98% 機率變有錢」。它是：在 default assumptions + default profile + exact ledger failure rules 下，10,000 個 lifetime paths 中約 69% 完成 funding plan。

第一個 audit 問題

如果你在 UI 完全沒改 default profile，FIRE screen 應重現 ~68.98%。差很多時，先查 path count、seed、profile state、tax region、spending、FIRE age。

第二個 learning point

90% 是「你的 target threshold」，confidence label 只是 view-layer wording。solver / curve / constraints 都看 90%，不看 Strong/High confidence 等 label。

FIRE Age Curve

把「退休年齡」從一個選定值，變成一條成功機率曲線。

03

FIRE Age Curve：它其實是把 Monte Carlo 重跑很多次

For each candidate age A: set targetFireAge = A → run full Monte Carlo → record success probability

不是插值：每個 whole age 都跑完整 lifetime simulation。

Common paths：每個 age 使用相同 seed/path indices，避免比較被 random luck 汙染。

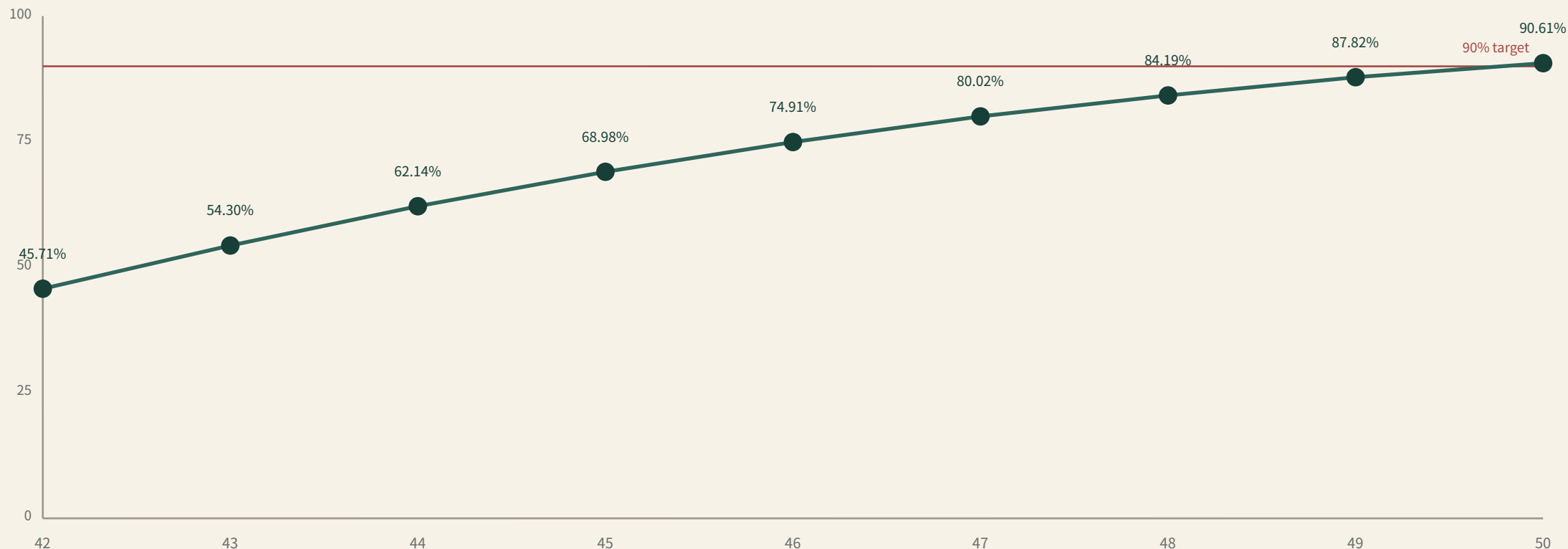
Output：curve + earliest age whose success probability clears your target.

Trade-off：延後 FIRE 同時增加 accumulation years、縮短 bridge/retirement years，通常會提高 success，但不是一個簡單 linear formula。

為什麼很有用

它比「你的 FIRE age 是 45」更 informative：你可以看到「多工作 1 年」到底買到多少 probability improvement。

Release age curve : 42 → 50 歲的 success probability



Earliest qualifying age = 50

45 → 50 : 68.98% → 90.61% (+21.63pp)

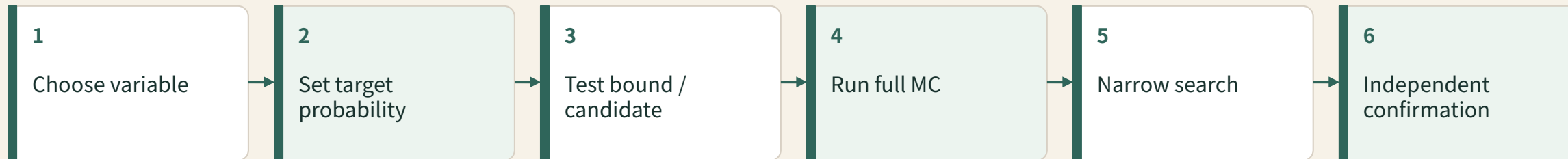
閱讀方式：curve 的斜率就是「多工作一年換到多少成功機率」；它把 salary growth、extra saving、shorter retirement、shorter bridge 全部一併反映。

Reverse Solver

把問題倒過來：不是「我的 plan 成功率多少」，而是「要改到多少才能達標？」

04

Reverse Solver : bounded search + complete simulation , 不是 algebra shortcut



Find smallest tested x such that $P_{\text{success}}(x) \geq \text{target}$

Answers are quantised to each mode's precision and re-run to confirm.

Common seed/path count candidate 間用同一市場路徑，讓 search 比較穩定。

Bounded 如果 target 在 supported bound 內達不到，就報「unreachable within bound」，不硬塞最大值。

Non-monotone aware FIRE age / pension contribution 不能盲信單調性；lock-up 可能讓 bridge 反而更差。

Search precision $\neq$ probability precision £100 salary precision 不代表 success probability 精確到 0.01%。

Reverse Solver 六種 mode：每一種「動的是什麼」都不同

Mode	方向	Search precision	真正修改的東西
Gross salary	increase	£100	salary only ; tax/NI/pension/lifestyle creep follow
Savings	increase	£100	working-life spending pound-for-pound 降低, 先 discretionary
FIRE age	increase	1 year	retirement start age
Retirement spending	decrease	£5 / month	只改 retirement spending ; working spending unchanged
Starting capital	increase	£500	explicit injection to selected account
Pension contribution	increase	0.5 percentage point	employee pension rate ; 可能因 liquidity 非單調

很重要：Salary solver 沒有同時幫你「最佳化 pension / spending / FIRE age」。它回答的是 ceteris paribus 的 required salary。

Worked example : 為什麼 required salary 是 £87,600

£87,600

smallest tested gross salary that clears 90%

£87,500

largest nearby non-clearing salary

90.02%

independent confirmation run

£87,500 → does not clear 90% | £87,600 → clears | confirm £87,600 → 90.02%

The release audit re-derived the same answer through Scenario Comparison and Reverse Solver.

這是「solve」不是比例放大

每個 candidate 都重新算： salary → salary sacrifice / pension → tax/NI → surplus → every future year → 10,000 MC paths → success.

但仍不是絕對真相

£87,600 對「目前 assumptions」成立；如果 spending、returns、FIRE age、tax policy 變，required salary 會變。

Solver 最容易誤解的 5 件事

1. £100 precision

只代表 search grid，不代表 finance truth 精確到 £100。

2. 90.02% vs 90.00%

10,000 paths 的 sampling error 仍約數十 basis points。

3. Bound

上界可因 pension allowance / unsupported plan 被縮小；是 model support boundary。

4. Evaluation budget

若 search 用完 budget，答案是 smallest qualifying tested value，不保證 mathematical minimum。

5. One variable at a time

它不是 multi-objective optimizer；salary solve 不會順便優化 pension / spending。

Marginal Allocation

真正回答最初的核心問題：下一筆 £1 / £1,000 到底該去哪裡？

05

Marginal Allocation：比較「同一筆增量」放到不同 destination



對每個 eligible destination：完整重跑 lifetime Monte Carlo

比較 output	為什麼重要
After-tax usable wealth at target ages	不是只看 nominal terminal balance
FIRE success + paired uncertainty	策略差異是否真的超過 sampling noise
Accessible capital	避免 pension 很高但 bridge 死掉
Lifetime tax	current relief vs future withdrawal tax
Debt / downside	property/mortgage 的槓桿風險
Eligibility reason	allowance、cash、property 缺失等不會被偷偷 cap

兩種 funding basis 完全不同: Gross earnings vs Existing after-tax cash

Gross earnings

把增量視為今年一次性的 non-pensionable bonus。
→ 重新算 exact incremental income tax + NI
→ pension 可透過 workplace method / match / NI shareback。

Existing after-tax cash

從 opening cash 移動已經是 after-tax 的錢。
→ ISA/GIA/cash 是 transfer
→ pension 走 RAS; provider relief + 可 claim relief 同年度 settlement / recycle。

Same “£1,000” amount $\neq$ same economic sacrifice under the two bases

所以 UI 必須清楚標示 basis; 不然 pension 看起來可能「憑空變大」。

Engine 明確避免 double credit: capital 不能既留在 cash 又被算成 pension/ISA。

不是「最高 expected wealth 就贏」：先過 constraint gates

FIRE probability

$\geq$ user target

Emergency cash

cash 不得掉到 reserve 以下

Minimum liquid years

accessible coverage 應達 threshold

Maximum debt

不得超過你設定的 real debt ceiling

Eligibility

ISA headroom / pension allowance / property exists / enough cash

Eligible candidates → rank by mean after-tax usable financial wealth at target ages

Locked pension and unsold property equity cannot fund the bridge.

No candidate passes?

Engine 應該回「no recommendation」，而不是硬排一個最不壞選項。這是設計上很重要的 honesty constraint。

Marginal result 怎麼讀：先看 feasibility，再看 economic ranking

Step	問題	你要避免的誤讀
1. Eligibility	這 destination 可以完整執行嗎？	不要把 silently capped contribution 當 full allocation
2. Constraints	是否維持 FIRE / liquidity / debt requirements ？	高 terminal wealth 不代表可退休
3. Paired delta	相對 baseline 的 wealth / success 差多少？	差異若落在 paired uncertainty $\approx$ tie
4. Tax path	當年 tax benefit vs future withdrawal tax ？	pension relief 不是 free money ； 有 lock-up + drawdown tax
5. Downside	worst/P10/debt exposure 有沒有惡化？	expected value 不能代表 risk appetite

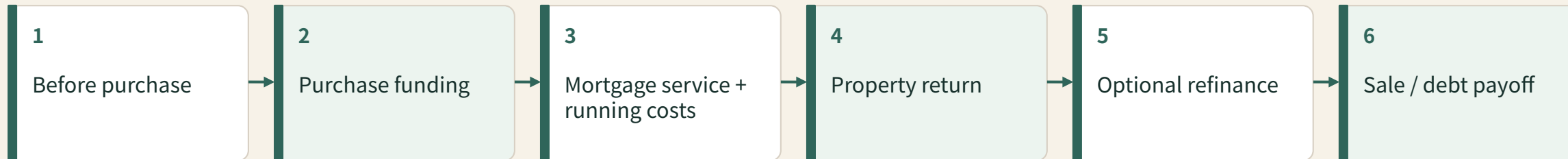
真正的「next £1」答案是 conditional：它取決於你當下 marginal tax band、ISA headroom、bridge liquidity、property debt、以及 FIRE constraint。

Property & Leverage

把房子當成 balance-sheet asset + debt + cash-flow obligation ，而不是只看「房價會漲」。

06

Property lifecycle : purchase / own / refinance / rent / sale 全進同一年度 ledger



Purchase price、deposit、legal costs、purchase tax 依 purchase age inflation；資金用正常 withdrawal order。

Atomic failure 整年 obligations fund 不到，就取消 purchase；不會出現「半棟房 + 已扣 deposit」。

Owner-occupied 從 household spending 移除已包含的 rent，但只在 owner occupation 期間。

Sale sale proceeds 先還 mortgage；negative equity 必須從其他資產 fund。

Equity access 只有 sale 會釋放 equity；沒有自動 cash-out refinance / forced liquidation。

Leverage : 房價報酬不是你的 equity 報酬

Property equity = property value - mortgage debt

LTV = mortgage debt / property value

例：£300k house / £60k deposit

初始 equity £60k , debt £240k , LTV 80% 。如果房價 +10% (忽略成本) , property gain £30k $\approx$ +50% of initial equity 。

同一槓桿也放大 downside

房價 - 10% $\rightarrow$ £30k value loss $\approx$ -50% of initial equity before interest / transaction costs 。

Economic property outcome $\approx$ appreciation + net rent - mortgage interest - operating costs - taxes - transaction costs

Principal repayment moves cash into equity; it is not the same economic cost as interest.

Owner-occupied vs Rental：現金流與 tax treatment 不一樣

Dimension	Owner occupied	Rental
Personal rent	included rent 從 spending 移除	不移除 personal rent
Gross rent income	none	occupancy-adjusted；management fee applies
Operating costs	household cash outflow	先算 operating profit / losses
Mortgage interest tax	personal cost	finance-cost restriction：20% tax reduction rules
Sale CGT	assumes full private residence relief	rental gain/loss 併入 GIA CGT + shared exemption/loss pool
Equity access	sale only	sale only

Audit 重点：不要把「mortgage principal」當 deductible rental expense；engine 不會。也不要把 gross rent 直接當 taxable profit。

Buy vs Rent+Invest : 真正公平的比較要處理 opportunity cost

Buy path

deposit + purchase tax + legal cost → less liquid capital
+ property appreciation
- interest / costs / selling costs
+ equity at sale



Rent + invest path

保留 opening financial assets + same spending schedule
在 purchase age , 最多把 avoided deposit / transaction costs
投資到 ISA/GIA (仍保 emergency reserve)

Compare on exact same market paths → terminal wealth, FIRE success, liquidity, drawdown, debt exposure

這個 screen 的價值不是「房子一定比較好 / 差」，而是把 leverage + liquidity + tax + opportunity cost 放回同一 lifetime objective。

Scenario Comparison

一次回答「如果我的收入、支出、 capital strategy 或整體 plan 不同，結果會怎樣？」

07

Scenario Comparison 有四種比較，不要混成一張表

1. 5×3×4 matrix

5 salary bands × 3 spending cases × 4 capital strategies = 60 complete simulations

2. Spending cases

£1,300 / £1,650 / £2,000 : 同時顯示 surplus side + FIRE capital side

3. Income bands

不同 gross salary 的 complete lifetime outcome

4. Named scenarios

Baseline / ISA Heavy / Pension Heavy / Income Growth / Property Heavy / No Property / Aggressive / Conservative FIRE

共通點：每一格都是完整 lifetime simulation ； 不是 spreadsheet-style local formula 。

5 × 3 × 4 matrix : 60 格到底在換什麼？

Salary axis

5 個 gross salary bands
(spec baseline includes £55k / £65k /
£75k / £90k / £120k)

Spending axis

Low / Base / High
預設 £1,300 / £1,650 / £2,000 per month

Strategy axis

ISA Heavy
Pension Tax-Band Optimised
Balanced
Property

5 salaries × 3 spending levels × 4 strategies = 60 full Monte Carlo cells

Full 10,000-path matrix measured ~271.72 seconds on the release hardware.

每個 cell 的 salary / spending / strategy 都會穿過同一 tax → ledger → Monte Carlo stack。

四個 capital strategy 並不是「名字」：它們是 explicit profile transforms

Strategy	做了什麼	沒做什麼
ISA Heavy	employee pension rate 降到保留 employer match 所需最低； surplus ISA→GIA；取消 planned property	不删除已持有 property
Pension Tax-Band Optimised	提高 workplace contribution，直到消除當前最高 occupied tax band；受 annual allowance / relevant earnings 限制	不是「max pension」；是 band-targeted
Balanced	保留你原本 contractual pension rate； surplus ISA→GIA；取消 planned purchase	沒有額外 tax-band optimisation
Property	使用 configured purchase： deposit/tax/mortgage/cost/rent replacement；其餘 surplus ISA→GIA	不是假設任意一棟 house

因此 Matrix 比較的是「具體策略實作」，不是抽象標籤。你改 scenario transform，就改了產品的 meaning。

8 個 named scenario preset：每一個都只做明確 transform

Preset	Transform
Baseline	profile unchanged
ISA Heavy	match-preserving pension minimum + ISA/GIA surplus
Pension Heavy	employee contribution +10pp， subject to allowance / earnings
Income Growth	salary 移到下一個 configured salary axis band
Property Heavy	使用 configured planned property purchase
No Property	移除 planned purchase；不能刪已持有 property
Aggressive FIRE	target FIRE age -3 years
Conservative FIRE	target FIRE age +3 years

好設計：如果 profile 根本無法表達某 preset，應該顯示原因，不要捏造一個看似可比較的 scenario。

Scenario cell 的 output：一格其實塞了兩層結果

Deterministic / current-year layer

gross income • take-home • income tax • employee NI • total tax • marginal rate + tax on next £1,000 • member/employer pension • ISA/GIA/cash contributions • investable surplus • spending • reference FIRE number

Simulation layer

success probability + SE • bridge/depletion probability • terminal mean/median/P10/worst • FIRE-age liquid/pension/property/net worth • fire-capital median • simulation count

為什麼需要兩層？

- Deterministic 告訴你「這個 plan 為什麼可投資 surplus 變多 / 變少」。
- Monte Carlo 告訴你「這些 cash-flow / wrapper changes 在 uncertainty 下是否真的改善 plan safety」。
- 只看 success probability，會看不到「是 salary、tax、spending 還是 allocation 在驅動」。

Spending sensitivity：支出同時打擊「今天能投多少」與「退休要多少」

Monthly spending	Annual retirement spend	Reference FIRE number @3.5%	FIRE success
£1,300	£15,600	£445,714	86.70%
£1,650	£19,800	£565,714	68.98%
£2,000	£24,000	£685,714	46.17%

£15,600 / 0.035 = £445,714

|

£19,800 / 0.035 = £565,714

|

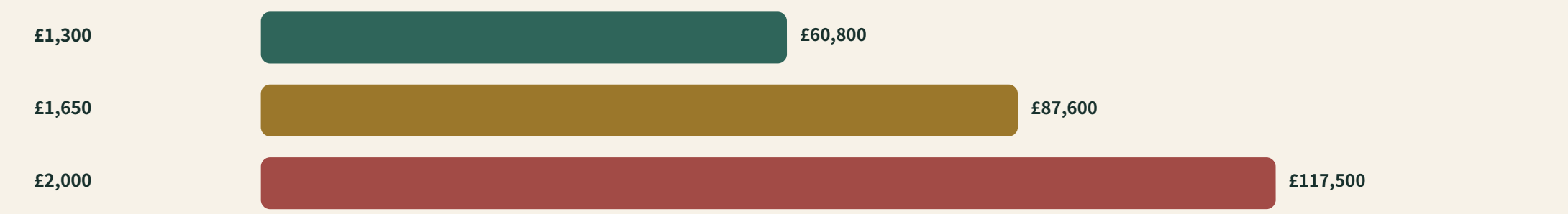
£24,000 / 0.035 = £685,714

Reference FIRE number arithmetic.



同三種支出， required salary 不是線性比例： 每格都真的 solve

Spending	Required salary @90%	Non-clear / Clear bracket	Confirmation	Full simulations
£1,300/mo	£60,800	£60,600 / £60,800	90.03%	24
£1,650/mo	£87,600	£87,500 / £87,600	90.02%	24
£2,000/mo	£117,500	£117,400 / £117,500	90.00%	23



Learning point：每多 £350/month 支出， required salary 的跳升不是固定 £X ，因為 tax bands、 pension contribution、 compounding 與 probability threshold 都是 nonlinear。

Scenario ranking：同一條 paths + sampling interval，避免把 noise 排成冠軍

$$\text{Paired delta} = \text{outcome_A}(\text{path } i) - \text{outcome_B}(\text{path } i)$$

If the lead sits inside paired sampling interval

報 tie / shared rank，而不是「A 贏 0.2pp」。

If differences clearly exceed uncertainty

才把它視為 model-evidenced lead。

這是很好的產品設計：Monte Carlo 是估計器，不是 deterministic scoreboard。

Audit 時，如果 UI 永遠硬排 1,2,3 而沒有 uncertainty / tie concept，就會過度解讀。

Where It Comes From

不要只問「結果是多少」；問「哪個 lever 真正把結果推上去 / 拉下來？」

08

Attribution interventions : 每一個 lever 都是完整 rerun

Income +£10k

salary + £10,000 / year

Expenses -£3k

all phases : discretionary first / floor respected

Band-optimised pension

pension rate → tax-band targeted rate

ISA → GIA

surplus ISA then GIA

GIA only

surplus 全部 GIA

FIRE +2 years

retire 2 years later

More equity risk

shift 10pp bonds/cash → equities

Property lever

+£10k deposit or shorten mortgage term by 5y

引擎只在 improvement 大於 sampling uncertainty 時，才把它當成 meaningful tested lever。

不要把 bars 加總成 100%

這不是 causal decomposition。每個 intervention 都從 baseline 各自改一件事重跑；effects 互相 overlap、nonlinear、不可相加。

九個 sensitivity family：專門測「assumption 錯了會怎樣」

Family	Low / High perturbation concept	它回答的問題
Equity return	mean $\pm$ 2pp	long-run return assumption 很敏感嗎?
Equity volatility	vol $\pm$ 5pp	same mean but more sequence risk?
Inflation	mean $\pm$ 1pp	real purchasing power / tax scaling sensitivity
Spending	configured low/base/high	lifestyle 是不是 dominant variable ?
Real salary growth	$\pm$ 1pp	active income path 的 importance
FIRE age	$\pm$ 2 years	timing lever 有多強?
Mortgage rate	$\pm$ 2pp	leverage 對 financing cost 的脆弱度
Property growth	$\pm$ 2pp	house-price assumption 影響多大
Pension withdrawal tax	+5pp / +10pp hypothetical surtax	future drawdown tax risk

每個 sensitivity case 仍然用 full path count + matching shocks。這是 scenario analysis，不是 confidence interval。

七個 deterministic stress：它們是「合成壓力路徑」，不是歷史回測機率

Stress	假設 path
2000-style crash	equities -15% / -20% / -25% over 3 years
2008-style crash	one-year equity -40%
High inflation decade	8% inflation for 10 years ; nominal asset means unchanged
Lost decade	equities 0% nominal for 10 years
Mortgage shock	9% mortgage rate for 3 years , recast remaining term
Property crash	property -30% once ; no automatic sale/equity release
Job loss	salary/bonus/pensionable pay/post-FIRE employment = 0 for 1 year ; then resumes

讀法

Stress 的用途是找「plan 會在這裡斷」，不是說 2008-style event 有 X% probability。看 failure age、funding shortfall、bridge liquidity 與 recovery。

「 Prediction 」到底是哪一種？ 這頁要牢記

Output	有沒有隨機性？	有沒有 probability meaning ?	可以說「預測」嗎？
Deterministic projection	No	No	最好叫 expected-value reference
Monte Carlo success	Yes	Yes, conditional on assumptions	可以叫 simulated probability estimate
Monte Carlo wealth percentile	Yes	Distribution conditional on assumptions	可以叫 modelled distribution
Sensitivity	Depends on MC underneath	No probability of assumption itself	if-X-then-Y analysis
Stress path	No (fixed synthetic path)	No	stress outcome only
Reverse solver	Uses MC repeatedly	Target crossing estimate	required value under current assumptions

最危險的語句：「模型預測我 50 歲退休」。更精確：在這組 assumptions 下，50 歲是 release baseline 中第一個 simulated success $\geq 90\%$ 的 whole age。

為什麼 Scenario / Solver 這麼慢？ 因為每格都是 full lifetime ledger × 10,000

Run	Release measured time	計算量概念
Single 10,000-path simulation	~4.3 s	10k lifetime ledgers
Property pairs	~5 s	paired complete plans
Marginal allocation	~25 s	multiple destinations × full MC
FIRE age curve 42–50	42.08 s	9 full MC runs
Where It Comes From	93.30 s	28 cases + 7 stresses
60-cell scenario matrix	271.72 s	60 full MC runs
3 full required-salary solves	290.84 s	~71 full MC evaluations
Full section61 harness	398 s	plus smokes / UI overhead

Engine 故意沒有偷偷降低 path count。真正 bottleneck 是 scalar year-by-year ledger；vectorisation 尚未做，因為最怕破壞 reconciliation identities。

已知 boundaries：不是 bug，但你必須知道模型沒有做什麼

Tax future

2026/27 structure constant-real；不是未來 legislated forecast

Expense shocks

stochastic expense shock 未實作；capital needs + 1y job-loss stress 是 deterministic half

Pension excess

annual allowance excess 不算 charge；candidate 變 infeasible / error，不 silently cap

Property

one residential property；無 cash-out refinance / automatic equity release / forced sale

GIA

aggregate-pool planning approximation；無 security-level 30-day matching

Solver

bounded、quantised、single-variable；不是 global optimiser

Runtime

path count 越高越慢；analysis cache 不跨 reload persisted

Advice

annual decision-support model；不是 payroll、tax return、regulated advice、guarantee

我會優先修的產品問題：不是 engine 全錯，而是「meaning 容易被 UI 混淆」

Priority	問題	建議修法
P0	Overview “Gross income” 容易被當 Gross salary	改成 Total gross income ； inline breakdown salary/interest/dividend/rent/pension withdrawal
P0	Facts / assumptions / technical knobs 混在同一長 sidebar	用 3 層 section + default collapse advanced model controls
P0	Scenario full run 很重	明確顯示 run cost 、 full vs preview badge 、 cache hit ； persist cache across reload
P1	MC probability 容易被當精確 truth	所有 probability 顯示 N 、 SE / interval 、 seed 、 assumption version
P1	Property output 容易只看 terminal wealth	同頁固定顯示 LTV 、 max debt 、 liquidity 、 bridge failures 、 transaction cost
P1	Reference FIRE number 易被當 safety target	UI 明確標「reference only」並跟 MC target visually separate
P2	Sensitivity / attribution 易被當 causal shares	copy 固定說 effects are non-additive / one-at-a-time interventions
P2	Tax-policy freeze 容易忘記	每個 long-horizon result 顯示 “2026/27 rules held constant in real terms”

當你覺得某個數字「怪怪的」：用這個 10-step audit

01

先截圖並記錄 profile inputs / active tab。

02

確認你看的是 salary 還是 total gross income。

03

用 first-year deterministic ledger 對 tax / NI / spending / pension。

04

檢查 asset balance：cash/ISA/GIA/pension/property/debt 是否分類正確。

05

檢查 event order：return 是否作用於 post-flow balance。

06

檢查 FIRE bridge：pension 是否被錯當 accessible。

07

MC 結果先確認 seed、path count、target probability。

08

比較 strategy 時確認 common paths / same assumptions。

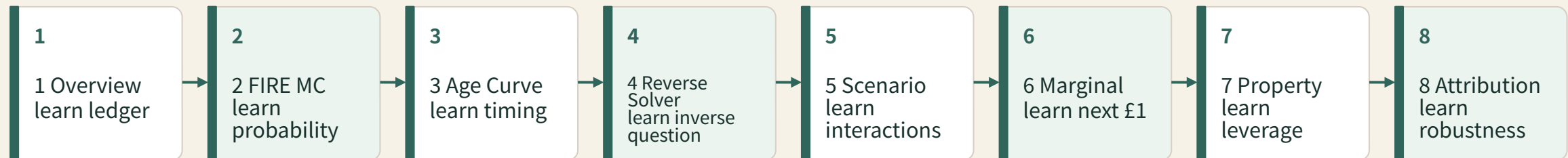
09

Solver 看 bracket、bound、confirmation，不只看 headline answer。

10

最後才去懷疑 stochastic model / market assumptions。

建議你真正「學會」這個模型的順序



我不建議從 Scenario Comparison 開始。它很有趣，但它是把前面所有 engine 組合在一起；若你還沒懂 ledger / MC，就只會看到一堆漂亮數字。

你現在最值得做的 practice

固定 default profile，不改 market assumptions。逐頁對照 release baselines：68.98%、age curve、£87.6k、spending cases。每一個都能重現後，再開始改你的個人 inputs。

Parameter appendix A : Personal • Income • Household • Spending

Parameter	Meaning / calculation effect
Current age	projection 起點; year t covers [age, age+1)
Target FIRE age	working → bridge boundary ; solver/curve main variable
End age	最後一個 funded year 之後的 boundary
Target success probability	pass threshold ; 不改 simulation path 本身
Gross salary	working years base salary
Bonus	employment income ; default non-pensionable
Other non-savings income	taxable non-savings income
Real salary growth	above inflation ; nominal salary compounds both
Post-FIRE employment	FIRE age onward ; not pensionable
State pension / age	from configured age ; also NI category C
Adults / children	household structure for supported spending breakdown
Current essential / discretionary	working-life household spend
Retirement essential / discretionary	default post-FIRE spend
Retirement floor / comfort	alternative spending runs
Included rent	removed only while owner-occupied
Lifestyle creep	share of real salary increases added to discretionary spend
Low / base / high	scenario monthly spending cases

Parameter appendix B : Assets • Pension • Wrappers • Liquidity

Parameter	Meaning / calculation effect
Cash	opening settlement/liquidity account
ISA	tax-free financial asset
GIA value / cost basis / losses	market value + CGT basis + carried losses
Workplace pension / SIPP	locked assets until access age
Pension tax-free cash used	reduces remaining lifetime lump-sum allowance
Employee / employer rate	recurring pension funding
Match up to / multiple	extra employer match based on employee rate
Employer NI shareback	salary-sacrifice NI saving credited to pension
Pension access age	before this, pension cannot fund spending
Carry-forward allowance	explicit verified amount; no inferred history
ISA allowance already used	reduces current-year subscription headroom
GIA dividend yield	opening GIA $\times$ yield $\rightarrow$ taxable dividend
GIA turnover / gain realisation	sell/rebuy activity $\rightarrow$ realised gain/basis step-up
Emergency reserve months	cash reserve gate based on essentials + required housing costs
Minimum liquid years	coverage diagnostic / marginal constraint
Capital needs	known one-off cash obligations at specified ages

Parameter appendix C : Portfolios • Market • Simulation

Parameter	Meaning / default
ISA / GIA / Pension weights	each wrapper equities+bonds+cash = 100%; default 85/10/5
Equity mean / vol	arithmetic nominal 7% / 16%
Bond mean / vol	3% / 6%
Cash mean / vol	2.5% / 0%
Property mean / vol	3.5% / 10%
Inflation mean / vol	2.5% / 0% in example
Correlation matrix	5×5 correlation of Gaussian log-growth shocks; default identity
Simulation paths	default 10,000 ; never auto-lowered
Random seed	421337 default ; shared paths depend on seed + path index
Reference withdrawal rate	3.5% ; only reference FIRE number
Withdrawal order	default cash → GIA → ISA → pension
Tax policy	constant_real ; 2026/27 structure scales with inflation

Parameter appendix D : Property 全欄位

Group	Parameters	Effect
Existing	value, mortgage balance, annual rate, remaining term	opening property/debt + amortisation
Running costs	maintenance, insurance, service charge, council tax	annual required housing cash flow
Rental	gross rent, occupancy, management fee	collected rent / taxable profit
Tax basis	acquisition cost basis, purchase tax override	rental CGT / manual tax cases
Purchase	age, price, deposit, legal/other costs	future acquisition commitment; amounts inflate to purchase age
Sale	age, selling-cost rate	sale before that year’ s operating flows/return
Refinance	age + new annual rate	remaining-term recast
Tax location/status	Scotland / England-NI / manual; standard / FTB / additional dwelling	purchase tax schedule
Use	owner-occupied / rental	rent replacement and tax treatment
Mortgage type	repayment / interest-only	principal schedule / final principal due

Source map：你要查「哪個數字從哪來」就看這裡

Question	Primary source
Input field definition	src/presentation/view/fields.ts
Default example assumptions	src/domain/fixtures.ts
Annual cash flow / gross income / withdrawals / failures	src/engine/ledger.ts
Tax / NI / pension / CGT	src/domain/tax/* + docs/tax-rules-2026-27.md
FIRE metrics	src/engine/fire-metrics.ts
Market path / portfolio returns	engine market/returns + simulation modules
Property / mortgage / tax	src/engine/property.ts + docs/property-model.md
Reverse search	src/engine/solver.ts
Scenario matrix / named presets	src/engine/scenario.ts
Marginal allocation	src/engine/marginal.ts + marginal-funding.ts
Attribution / sensitivity	src/engine/attribution.ts
Stress assumptions	src/engine/stress.ts
Verified release benchmark	docs/handoffs/chunk-10.md
Product authority	docs/lifetime-capital-allocation-fire-optimisation-spec-v0.3.md

一頁 cheat sheet：每個 screen 只記這一個問題

Overview

我的錢現在怎麼流？

FIRE & MC

這個 plan 在 uncertainty 下多常活到最後？

Age Curve

多工作一年值多少成功機率？

Reverse Solver

我要改多少才跨過 target？

Marginal

下一筆錢放哪裡最有效？

Property

槓桿後，房子真的改善 lifetime plan 嗎？

Scenario

不同收入 / 支出 / 策略組合，哪個比較好？

Where It Comes From

真正驅動結果的是什麼？假設錯了會不會翻盤？

你真正要學會的不是 8 個畫面，而是一條因果鏈：

Input → Tax → Annual Ledger → Market Path → Lifetime Outcome → Comparison / Search → Decision